

# Action-Conditioned Predictive Consistency as a Diagnostic for Gaussian-Noise Robustness in JEPA World Models

Author names to be supplied for arXiv v1

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## Abstract

Latent predictive world models avoid pixel reconstruction, but pixel-free prediction is not a robustness definition for closed-loop control. We study this gap with action-conditioned predictive consistency (ACPC), a no-retraining diagnostic for JEPA-style control checkpoints. The compressed paper-facing diagnostic has two theory-aligned readouts: ACPC tail risk (ATR), the high-quantile same-state clean/noisy action-conditioned rollout disagreement, and selective-margin pass rate (SMPR), a task-grounded near-boundary check that action-relevant distinctions remain separated. Across three independent LeWM training seeds and four control tasks, no-noise checkpoints are fragile under observation-only Gaussian noise, while input-side noise training yields broad matched-Gaussian robustness plateaus. At recovered endpoints, ATR drops sharply and SMPR rises to near one across tasks. The claim is a controlled diagnostic for matched-Gaussian robustness plateaus, not a closed-loop stability guarantee, universal perturbation-transfer result, or new training objective.

**Keywords:** visual world models; JEPA; action-conditioned predictive consistency; visual robustness; predictive-dynamics diagnostics.

**Code and data release:** <https://github.com/Anguo-star/lewm-acpc-diagnostics>

## 1 Introduction

### 1.1 Latent prediction is not a robustness definition

JEPAs predict future representations instead of reconstructing pixels, which can reduce pressure to encode high-magnitude irrelevant visual detail [1, 3, 29]. For control, however, “irrelevant” is action- and transition-dependent. A representation may ignore a background texture but must preserve a contact pose or doorway state if that changes the next useful action. Thus encoder-level invariance is an incomplete target: planning uses the model’s *predictions*, so robustness must be stated after action-conditioned prediction.

We use this failure as a controlled diagnostic probe. Across three independent LeWM [19] training seeds, the no-noise baseline remains strong on clean observations but loses  $74.56 \pm 5.55$  points on PushT and  $41.00 \pm 1.44$  points on Reacher under observation-only Gaussian noise at std 0.08 with the goal unperturbed. Observation+goal noise is reported as a harder auxiliary stress condition. Test-time visual fragility is not new [15, 33, 12, 21]; our focus is what it reveals about JEPA latent world-model control.

### 1.2 Operationalizing action-conditioned predictive consistency

We operationalize a downstream-control view of visual robustness for JEPA world models through *action-conditioned predictive dynamics*. Let  $z = E(h)$  and  $\tilde{z} = E(\tilde{h})$  be the latents an encoder  $E$  produces from an unperturbed history  $h$  and a perturbed history  $\tilde{h}$  of the same underlying state, and let  $F^k(\cdot, \mathbf{a}_{t:t+k-1})$  be the action-conditioned latent predictor rolled out for  $k$  steps under an action sequence  $\mathbf{a}$ . We *allow*  $z \neq \tilde{z}$ . The robustness target is placed *after* the predictor: in task-relevant coordinates,

$$F^k(z, \mathbf{a}_{t:t+k-1}) \approx F^k(\tilde{z}, \mathbf{a}_{t:t+k-1}), \quad k = 1, \dots, H, \quad (1)$$

so that the two branches induce the same next-state and rollout predictions under the same action intervention. The requirement is *selective*: it should hold only for nuisance perturbations of the same underlying state. State

differences that change action-conditioned transitions, costs, or the optimal action must remain *discriminable*, or the planner loses the distinctions it needs. We make both halves precise in Section 3.

This diagnostic view changes what counts as a useful robustness measurement. Encoder geometry is a useful first-stage diagnostic because it shows whether visual perturbations enter and reshape the latent neighborhood, but encoder-level latent closeness,  $\|z - \tilde{z}\| \rightarrow 0$ , is not a complete control-robustness definition: large nuisance shifts can wash out after prediction, whereas small shifts can still be amplified downstream. Planning, cost, and action metrics remain useful as *downstream readouts* of predictive consistency; future objectives should regularize action-conditioned predictive dynamics while preserving action-relevant discriminability.

### 1.3 Existing noisy-evaluation results as a controlled probe

We use input-side noise augmentation as the controlled intervention. The  $\sigma_{\max} \in \{0.01, \dots, 0.08\}$  sweep across four tasks creates paired baseline/recovered checkpoints for asking whether closed-loop Gaussian recovery is accompanied by lower same-state predictive tail risk and preserved task-grounded separations. Because the sweep forms broad task-dependent plateaus under finite evaluation variance, we do not treat a single point-best  $\sigma_{\max}$  or small within-plateau score differences as meaningful checkpoint ordering.

### 1.4 Contributions

**C1 — Selective ACPC principle.** Robustness diagnostics should require both low same-state action-conditioned predictive-disagreement tails and preserved task-grounded near-boundary separations. Low consistency loss alone is not enough, because collapse can also make noisy and clean views agree.

**C2 — Theory-matched compression.** ATR estimates the ACPC-tail term exposed by the sampled-pool analysis and local Gaussian sensitivity calculation. SMPR estimates the discriminability/no-collapse countercondition that makes low ATR meaningful. The clean-margin term remains a formal caveat explaining why the diagnostic is not a closed-loop guarantee.

**C3 — Fixed-checkpoint evidence.** Across three LeWM training seeds, matched-Gaussian robustness plateaus coincide with large ATR reductions and near-one SMPR at recovered endpoints. Unseen-stressor scores bound the perturbation scope without turning the diagnostic into a method or universal transfer claim.

## 2 Related Work

### 2.1 Latent prediction and JEPA world models

JEPA [16] predicts in latent space rather than reconstructing pixels; I-JEPA [1] and the V-JEPA family [3, 2, 20] extend this idea to images, video, and dense physical-world representations. LeWM [19], our baseline, stabilizes an end-to-end JEPA world model with SIGReg [6]. PLDM [25, 26], via `stable-worldmodel` [18], provides related latent-dynamics planning context. LeJEPA theory [14] studies latent-variable recovery and latent planning; our diagnostic asks whether learned visual world-model checkpoints give paired clean/noised views consistent action-conditioned predictions.

### 2.2 Invariance, augmentation, and visual robustness

Augmentation and joint-embedding losses impose invariances only when their training signal identifies the right nuisance factors [31, 27, 35]. Recent theory clarifies why latent prediction can help with nuisance or noisy features but also why it is not enough for control. Van Assel et al. [29] and Littwin et al. [17] analyze representation-level effects; they do not address closed-loop action-conditioned rollout, candidate-cost stability, or the need to keep action-distinct states separable. Seq-JEPA [8] addresses a related invariance–equivariance tension architecturally.

We do not train robust visual-control systems as competing baselines. DrQ/DrQ-v2 [15, 33], SODA/DMC-GB [12], DreamerV3 [10], TD-MPC2 [11], and ViGMO [21] define future method-comparison axes; noisy/video JEPA variants [22, 13, 23] are representation-level context. ACPC is complementary: it tests whether same-state visual perturbations become equivalent after a fixed action intervention while action-, transition-, or cost-distinct cases remain discriminable.

### 2.3 Control-relevant predictive representations

Bisimulation merges states with equivalent reward and transition behavior. DeepMDP [7], DBC [34], bisimulation-regularized MPC [24], and Bisim-JEPA [28] use this principle to learn control-relevant representations. Value-equivalent and value-aware models [9, 30] and VIBR [5] likewise emphasize downstream control consequences. Wang et al. [32] formalize action-conditioned video world modeling as a group action with identity, inverse, and composition-consistency metrics. ReOI [4] treats visual distractor robustness in visual MPC through observation interventions for world-model action-outcome prediction. These lines show that downstream predictive consequences are already central to control-representation work. Our diagnostic instantiates that view for paired clean/noised JEPA latent-world-model rollouts: same-state noised and unperturbed views should agree after the same action sequence, while action-distinct transitions remain separable.

Our use of ACPC is therefore narrower than an identifiability theorem, a learned bisimulation metric, or a robust MPC intervention. LeJEPA-style results motivate latent prediction and planning under latent recovery assumptions; bisimulation and value-aware models define task abstractions; group-action metrics test action faithfulness; ReOI modifies observation handling for robust visual MPC. This paper instead supplies a JEPA checkpoint diagnostic centered on additive Gaussian pixel-noise stress tests: predictive stability is tested on paired same-state clean/noised rollouts under the same action sequence, fixed-candidate cost stability is derived as a sufficient condition, and a separate discriminability guard checks that action-, transition-, or cost-distinct cases have not been collapsed.

## 3 Action-Conditioned Predictive Consistency

This section defines action-conditioned predictive consistency and its discriminability countercondition. We use ACPC to separate two requirements that encoder invariance alone conflates: same-state visual perturbations should agree after action-conditioned rollout, while task-grounded different-state pairs should remain separable. Section 3.2 then maps these two requirements to ATR and SMPR.

### 3.1 Setup and definitions

Let  $o_t$  be an unperturbed observation and  $\tilde{o}_t = \tau(o_t)$  a visually perturbed view of the *same* underlying state ( $\tau$  is additive Gaussian pixel noise here, but the definitions are perturbation-agnostic). Let  $h_t$  and  $\tilde{h}_t$  be the corresponding observation-action histories,  $E_\theta$  the encoder, and  $F_\theta$  the action-conditioned latent predictor. Fix an action sequence  $\mathbf{a}_t^H = a_{t:t+H-1}$  that is *identical* on both branches. For compact notation below,  $\mathbf{a}_{0:k-1}$  denotes the length- $k$  prefix of this sequence after re-indexing time at  $t$ , and we write

$$z_t = E_\theta(h_t), \quad \tilde{z}_t = E_\theta(\tilde{h}_t), \quad \hat{z}_{t+k} = F_\theta^k(z_t, \mathbf{a}_{0:k-1}), \quad \hat{\tilde{z}}_{t+k} = F_\theta^k(\tilde{z}_t, \mathbf{a}_{0:k-1}). \quad (2)$$

Let  $\Pi$  denote a *task-relevant predictive readout* — the full latent, a transition delta, goal-distance or cost features, or a learned projection — and let  $d$  be a distance on its range.

**Action-conditioned predictive consistency.** For a same-state perturbation pair under the shared action sequence,

$$\text{ACPC}_H(o_t, \tilde{o}_t, \mathbf{a}) = \sum_{k=1}^H \alpha_k d(\Pi(\hat{z}_{t+k}), \Pi(\hat{\tilde{z}}_{t+k})), \quad \alpha_k \geq 0. \quad (3)$$

For the theoretical statements below, we use the induced horizon metric

$$d_H((u_1, \dots, u_H), (v_1, \dots, v_H)) = \sum_{k=1}^H \alpha_k d(u_k, v_k), \quad (4)$$

so that Equation (3) is exactly the clean/perturbed rollout-readout distance under the shared action sequence. In experiments, ATR uses the 90th percentile of this normalized same-state rollout disagreement. In this diagnostic framing, robustness corresponds to small  $\text{ACPC}_H$  together with action-relevant discriminability, not to the encoder distance  $d(z_t, \tilde{z}_t)$  alone. Encoder closeness is a useful first-stage risk signal, but it is neither necessary (a large but task-irrelevant encoder shift can wash out through  $F_\theta$  and  $\Pi$ ) nor sufficient (a small encoder shift can still change the predicted future enough to flip the planned action).

**Action-relevant discriminability (countercondition).** Consistency must be *selective*. Let  $Y^H(s, \mathbf{a})$  denote a task-relevant future readout that is external to the learned predictor — simulator state, cost-to-go, inverse-dynamics label, contact/keyframe tag, or another diagnostic proxy available in the dataset. For two genuinely different states  $s_i, s_j$  (with latents  $z_i, z_j$ ) that admit an action sequence under which this external future readout diverges,

$$\Delta_{\text{dyn}}^H(s_i, s_j, \mathbf{a}) = d_Y(Y^H(s_i, \mathbf{a}), Y^H(s_j, \mathbf{a})) > m, \quad (5)$$

the representation and predictions should keep them separable,

$$d(\Psi(z_i), \Psi(z_j)) > m', \quad (6)$$

where  $\Psi$  is a discriminability readout, possibly action-conditioned (latent, predicted delta under  $\mathbf{a}$ , inverse-dynamics feature, cost feature, or rollout embedding), and  $m, m' > 0$  are margins. Equations (3)–(6) state the target: lower predictive disagreement for same-state visual perturbations must preserve discriminability among state/action/transition-distinct pairs. A method that drove  $\text{ACPC}_H \rightarrow 0$  by collapsing the latent would violate (6), so the two conditions are inseparable.

### 3.2 Operational consistency diagnostics

The paper-facing diagnostic uses two readouts. ACPC Tail Risk (ATR) is the 90th percentile of normalized same-state clean/noisy ACPC-H rollout disagreement under a fixed recorded action sequence; lower is better. Selective Margin Pass Rate (SMPR) is the pass rate of task-grounded label-crossing pairs whose clean different-state rollout distance exceeds the same-state noisy radius at margin zero; higher is better. ATR measures predictive-stability tail risk, while SMPR rules out the collapse failure in which all states become indistinguishable. Encoder geometry and planner-side readouts are not separate paper-facing diagnostics in this compressed version.

**Table 1:** Theory-to-metric mapping for compressed selective ACPC.

| Core theorem quantity                   | Paper-facing readout | Role                                                    |
|-----------------------------------------|----------------------|---------------------------------------------------------|
| $\Pr[D > \epsilon] / \text{ACPC tail}$  | ATR                  | same-state action-conditioned predictive-stability tail |
| $d_{\text{diff}} > d_{\text{same}} + m$ | SMPR                 | task-grounded anti-collapse margin                      |

### 3.3 ACPC as a fixed-candidate stability condition

ACPC is connected to planning through the candidate costs evaluated by model-predictive control. Let  $h$  be a clean history,  $\tilde{h}$  the corresponding perturbed history, and let both branches evaluate the same candidate action sequence  $\mathbf{a}$ . For a goal or task descriptor  $g$ , define

$$C_h(\mathbf{a}, g) = J(\Pi(\hat{z}_{t+1}), \dots, \Pi(\hat{z}_{t+H}), g), \quad (7)$$

where  $J$  is the planner’s cost readout on the rollout-readout sequence.  $C_{\tilde{h}}(\mathbf{a}, g)$  is defined analogously from the perturbed branch. The following statements are sufficient conditions for stability on a fixed candidate set; they do not prove stability of the full CEM sampling distribution or of the closed-loop trajectory over repeated replanning.

**Proposition 1** (Encoder shift as one route to rollout disagreement). *Fix an action sequence  $\mathbf{a}$  and write*

$$G_{\mathbf{a}}(z) = \Pi(F_{\theta}^{1:H}(z, \mathbf{a}))$$

*for the rollout-readout map. If  $G_{\mathbf{a}}$  is locally  $L_G$ -Lipschitz between  $E_{\theta}(h)$  and  $E_{\theta}(\tilde{h})$  under encoder metric  $d_Z$  and rollout-readout metric  $d_H$ , then*

$$d_H(G_{\mathbf{a}}(E_{\theta}(h)), G_{\mathbf{a}}(E_{\theta}(\tilde{h}))) \leq L_G d_Z(E_{\theta}(h), E_{\theta}(\tilde{h})).$$

*Proof.* This is the local Lipschitz condition applied to the two encoded histories.  $\square$

This bound makes encoder geometry a first-stage risk signal: small encoder shifts imply small rollout disagreement only under local insensitivity, large shifts can be contracted by  $G_{\mathbf{a}}$ , and small shifts can be amplified by the predictor or cost readout. This is why the empirical diagnostic measures the tail of action-conditioned rollout disagreement rather than encoder distance alone.

**Proposition 2** (ACPC controls candidate-cost drift). *Assume that  $J$  is locally  $L_J$ -Lipschitz on the clean/perturbed rollout-readout neighborhood evaluated by the fixed candidate set under metric  $d_H$ . If*

$$d_H(\Pi(F_\theta^{1:H}(E_\theta(h), \mathbf{a})), \Pi(F_\theta^{1:H}(E_\theta(\tilde{h}), \mathbf{a}))) \leq \epsilon,$$

then

$$|C_h(\mathbf{a}, g) - C_{\tilde{h}}(\mathbf{a}, g)| \leq L_J \epsilon.$$

*Proof.* The inequality is the local Lipschitz condition applied directly to the clean and perturbed rollout-readout sequences.  $\square$

In the compressed diagnostic, this condition is instantiated by the L2 rollout-token distance whose high-quantile tail defines ATR; empirical companion audits are kept outside the paper-facing metric set.

**Proposition 3** (Candidate top-1 stability). *Let  $\mathcal{A} = \{\mathbf{a}^1, \dots, \mathbf{a}^K\}$  be the shared candidate set. Assume lower cost is preferred, and let  $\mathbf{a}^{(1)}$  and  $\mathbf{a}^{(2)}$  denote the lowest- and second-lowest-cost candidates on the clean branch. Suppose that for every candidate  $j$ ,*

$$|C_h(\mathbf{a}^j, g) - C_{\tilde{h}}(\mathbf{a}^j, g)| \leq \eta.$$

*If the clean branch has a strict top-1/top-2 margin*

$$\Delta = C_h(\mathbf{a}^{(2)}, g) - C_h(\mathbf{a}^{(1)}, g) > 2\eta,$$

*then the perturbed branch selects the same top-1 candidate.*

*Proof.* For any candidate  $\mathbf{a}^j \neq \mathbf{a}^{(1)}$ ,

$$C_{\tilde{h}}(\mathbf{a}^j, g) - C_{\tilde{h}}(\mathbf{a}^{(1)}, g) \geq C_h(\mathbf{a}^j, g) - C_h(\mathbf{a}^{(1)}, g) - 2\eta \geq \Delta - 2\eta > 0.$$

Thus every non-top candidate remains more costly than the clean top-1 candidate on the perturbed branch.  $\square$

**Corollary 1** (ACPC-margin condition for candidate stability). *Let  $\mathcal{A} = \{\mathbf{a}^1, \dots, \mathbf{a}^K\}$  be a shared candidate set and assume the cost readout  $J$  is locally  $L_J$ -Lipschitz as in Proposition 2. If every candidate in  $\mathcal{A}$  has rollout-readout discrepancy at most  $\epsilon$  between the clean and perturbed branches, and the clean branch has top-1/top-2 margin  $\Delta > 2L_J\epsilon$ , then the two branches select the same top-1 candidate from  $\mathcal{A}$ .*

*Proof.* Proposition 2 gives  $|C_h(\mathbf{a}^j, g) - C_{\tilde{h}}(\mathbf{a}^j, g)| \leq L_J\epsilon$  for every candidate. Applying Proposition 3 with  $\eta = L_J\epsilon$  gives the result.  $\square$

The corollary gives a formal link between ACPC and fixed-candidate decisions without turning the diagnostic into a closed-loop guarantee. It connects paired predictive agreement to candidate-cost stability and then to action selection under a margin condition. The result is limited to a fixed candidate set; adaptive sampling, repeated replanning, and environment feedback can still change the closed-loop trajectory.

### 3.4 Sampled-pool stability and Gaussian sensitivity

The fixed-candidate statement is the deterministic core. A sampled candidate pool adds finite-sample risk. The next bound keeps the same diagnostic scope while exposing why ATR must be a tail readout and why clean candidate margins remain a caveat.

**Theorem 1** (Sampled-pool ACPC stability). *Let  $\mathcal{A} = \{\mathbf{a}^1, \dots, \mathbf{a}^K\} \sim q^K$  be a once-sampled candidate pool, and use deterministic tie-breaking for  $\arg \min$ . For each sampled candidate define*

$$D_j = d_H(\Pi(F_\theta^{1:H}(E_\theta(h), \mathbf{a}^j)), \Pi(F_\theta^{1:H}(E_\theta(\tilde{h}), \mathbf{a}^j))).$$

*Assume  $J$  is locally  $L_J$ -Lipschitz on the evaluated rollout-readout neighborhood and that a single draw from  $q$  satisfies*

$$\Pr_{\mathbf{a} \sim q}[D(\mathbf{a}) > \epsilon] \leq \delta.$$

*Let  $\Delta_{\mathcal{A}} = C_h(\mathbf{a}^{(2)}, g) - C_h(\mathbf{a}^{(1)}, g)$  be the clean branch top-1/top-2 margin inside the sampled pool. Then*

$$\Pr_{\mathcal{A}} \left[ \arg \min_j C_h(\mathbf{a}^j, g) \neq \arg \min_j C_{\tilde{h}}(\mathbf{a}^j, g) \right] \leq K\delta + \Pr_{\mathcal{A}}[\Delta_{\mathcal{A}} \leq 2L_J\epsilon].$$

Theorem 1 is not a convergence theorem for adaptive planning. It controls the instability of one sampled candidate pool. The first term is the probability that at least one candidate has a large clean/noisy rollout discrepancy, which motivates ATR. The second term is the probability that the clean candidate pool has too small a top-1 margin; it explains why ATR is not a closed-loop guarantee. SMPR is separate from this flip bound: it is the selective-discriminability condition required to make low ATR meaningful rather than collapsed.

**Corollary 2** (Replanning union bound). *If replanning step  $t$  has sampled-pool flip-probability bound*

$$r_t = K_t \delta_t + \Pr[\Delta_{\mathcal{A}_t} \leq 2L_J \epsilon_t],$$

*then over  $T$  replanning steps,*

$$\Pr[\exists t \leq T : \text{the sampled-pool top-1 flips at step } t] \leq \sum_{t=1}^T r_t.$$

**Proposition 4** (Finite-sample tail calibration for paired diagnostics). *Fix the diagnostic sampling distribution used to draw independent paired clean/noisy histories and candidate action sequences. Let*

$$X_i = \mathbf{1}[D_i > \epsilon], \quad i = 1, \dots, n,$$

*where  $D_i$  is the paired ACPC rollout-readout discrepancy for diagnostic item  $i$ . Let  $p_\epsilon = \Pr[D > \epsilon]$  and  $\hat{p}_\epsilon = n^{-1} \sum_i X_i$ . Then, with probability at least  $1 - \delta$  over the diagnostic sample,*

$$p_\epsilon \leq \hat{p}_\epsilon + \sqrt{\frac{\log(1/\delta)}{2n}}.$$

*The same one-sided calibration applies to SMPR after fixing its task-grounded pair construction.*

This calibration is with respect to the fixed diagnostic sampling distribution and its independent-sample idealization. It does not convert an empirical quantile or pass-rate table into a uniform bound over all closed-loop states; when nearby dataset windows are dependent, we use it only as calibration.

The Gaussian stressor also has a local sensitivity interpretation. Let  $G_{\mathbf{a}}(z) = \Pi(F_\theta^{1:H}(z, \mathbf{a}))$  and let  $\tilde{o} = o + \xi$  with  $\xi \sim \mathcal{N}(0, \sigma^2 I)$ .

**Proposition 5** (Local Gaussian ACPC sensitivity). *If  $E_\theta$  and  $G_{\mathbf{a}}$  are locally twice differentiable around  $o$  and  $E_\theta(o)$  with bounded second-order remainders, then for small  $\sigma$ ,*

$$G_{\mathbf{a}}(E_\theta(o + \xi)) - G_{\mathbf{a}}(E_\theta(o)) = J_{G_{\mathbf{a}}}(E_\theta(o))J_E(o)\xi + R_\xi,$$

*where the remainder contributes higher-order terms, and*

$$\mathbb{E}_\xi \left[ \|G_{\mathbf{a}}(E_\theta(o + \xi)) - G_{\mathbf{a}}(E_\theta(o))\|_2^2 \right] = \sigma^2 \|J_{G_{\mathbf{a}}}(E_\theta(o))J_E(o)\|_F^2 + O(\sigma^3).$$

Thus Gaussian ACPC measures action-conditioned encoder–predictor sensitivity, not encoder invariance alone. Small encoder shifts can still be amplified by  $J_{G_{\mathbf{a}}}$ , while large nuisance shifts can become harmless after rollout if the product  $J_{G_{\mathbf{a}}}J_E$  is small in those directions. The selective target is therefore asymmetric: the product should be small along nuisance perturbation directions, while rollout readouts for action-relevant state differences remain separated. The task-grounded near-boundary proxy margin pass event used later is

$$M = \mathbf{1}[d_{\text{proxy diff}} > d_{\text{same state noise}} + \delta_m],$$

with  $\delta_m = 0$  in the reported table. It is a finite state-proxy selective-consistency check, not a full semantic-label coverage guarantee.

The theoretical statements in this section are intended as diagnostic calibration rather than as closed-loop control guarantees. They identify the failure modes that a paired clean/noisy diagnostic should measure: rollout-disagreement tails, clean candidate-margin caveats, and collapse of task-grounded separations. They do not assert that a low empirical ATR is a uniform bound for adaptive planning, repeated replanning, or environment-feedback stability.

**Selective ACPC pseudo-metric.** For a nonempty fixed candidate family  $\mathcal{A}$ , the stability condition can be summarized by

$$D_{\text{ACPC}}^{\mathcal{A}}(h, \tilde{h}) = \max_{\mathbf{a} \in \mathcal{A}} d_H(\Pi(F_{\theta}^{1:H}(E_{\theta}(h), \mathbf{a})), \Pi(F_{\theta}^{1:H}(E_{\theta}(\tilde{h}), \mathbf{a}))). \quad (8)$$

When  $d_H$  is a metric (or a pseudo-metric, if some horizon weights vanish),  $D_{\text{ACPC}}^{\mathcal{A}}$  is a pseudo-metric on histories after passing through the encoder, predictor, and readout: distinct histories can have zero distance if all candidate rollouts are readout-equivalent. This follows because each map  $h \mapsto \Pi(F_{\theta}^{1:H}(E_{\theta}(h), \mathbf{a}))$  pulls back  $d_H$  to a pseudo-metric, and the pointwise maximum of pseudo-metrics is again a pseudo-metric. The useful object is therefore *selective* predictive stability. Small  $D_{\text{ACPC}}^{\mathcal{A}}$  for same-state visual perturbations gives candidate-cost and top-1 stability under the preceding Lipschitz and margin assumptions; the same contraction is not desirable for state/action pairs whose external task readouts differ, which is why the discriminability guard is part of the definition rather than an auxiliary aesthetic check.

**Corollary 3** (Fixed-candidate predictive stability). *Assume  $D_{\text{ACPC}}^{\mathcal{A}}(h, \tilde{h}) \leq \epsilon$  for a same-state visual-perturbation pair and that  $J$  is locally  $L_J$ -Lipschitz on the rollout readout. If the clean candidate margin satisfies  $\Delta > 2L_J\epsilon$ , then the clean and perturbed branches select the same top-1 candidate from the fixed set  $\mathcal{A}$ .*

*Proof.* By definition of  $D_{\text{ACPC}}^{\mathcal{A}}$ , every candidate in  $\mathcal{A}$  has rollout-readout discrepancy at most  $\epsilon$ . Proposition 2 bounds the cost drift of every candidate by  $L_J\epsilon$ , and Proposition 3 gives top-1 stability when  $\Delta > 2L_J\epsilon$ .  $\square$

The statement becomes selective only when it is paired with a discriminability condition such as Equation (6). Without that second condition, fixed-candidate predictive stability can still be achieved by a collapsed encoder–predictor, as shown next.

Encoder closeness is not the right substitute for this condition. It is not necessary: a nuisance-subspace change can move  $E_{\theta}(h)$  and  $E_{\theta}(\tilde{h})$  far apart while leaving  $\Pi(F_{\theta}^{1:H}(\cdot, \mathbf{a}))$  and the cost unchanged. It is not sufficient either: a small encoder displacement can flip a low-margin cost ranking when the predictor or cost readout has high local sensitivity.

**Proposition 6** (ACPC alone permits collapse). *There exist encoder–predictor pairs for which  $\text{ACPC}_H = 0$  for every same-state perturbation pair and every action sequence, while action-distinct states are mapped to identical rollout readouts.*

*Proof.* Let  $E_{\theta}(h) = z_0$  for every history and let  $F_{\theta}^k(z_0, \mathbf{a}) = u_0$  for every horizon  $k$  and action sequence  $\mathbf{a}$ . Then clean and perturbed branches always have identical rollout readouts, so same-state ACPC is zero. However, any two states that require different actions, costs, or transitions also have identical readouts, violating the discriminability condition in Equation (6).  $\square$

Thus low ACPC is meaningful only with a guard on action-relevant separation. This is the role of SMPR in the paper-facing diagnostic. This is where ACPC differs from existing nuisance-feature analyses of latent prediction: those results help explain why latent prediction may suppress irrelevant or noisy features at the representation level, whereas the condition above asks whether the remaining perturbation changes action-conditioned rollout readouts while preserving task-grounded separations.

## 4 Study protocol

### 4.1 Models and noise intervention

LeWM [19] is an end-to-end JEPA world model trained with a latent prediction loss and SIGReg regularization for latent-space model-predictive control. PushT (2D pushing), TwoRoom (2D navigation), Reacher (2D arm), and Cube (3D manipulation) form a heterogeneous stress panel spanning different nuisance/discriminability regimes. We add per-frame Gaussian noise to the input sequence as a full-sequence augmentation: each frame is independently noised with probability  $p = 1.0$ , and the standard deviation is drawn from  $\text{Uniform}(0, \sigma_{\max})$ . This is a full-sequence input-side augmentation. We sweep  $\sigma_{\max} \in \{0.01, \dots, 0.08\}$  and evaluate observation-only Gaussian noise with the goal image unperturbed as the primary robustness endpoint. Observation+goal Gaussian noise is retained as an auxiliary stress condition.

## 4.2 Evaluation and diagnostics

The canonical LeWM development grid contains 36 epoch-10 checkpoints for training seed 3072: four tasks and, for each task, a no-noise baseline plus eight noise-trained checkpoints. Each checkpoint is evaluated with three evaluation seeds (42/43/44) and 100 trajectories per seed; canonical tables report the population mean  $\pm$  population std across those three evaluation seeds. A completed Gaussian training-seed replication adds independent LeWM training seeds 3073 and 3074 under the same nine-point Gaussian sweep and evaluation protocol. Together, seed 3072 plus replication seeds 3073/3074 provide three training seeds for the main Gaussian closed-loop endpoint in Table 4.

The main results follow the compressed evidence chain used in the claims. Closed-loop tables report training-seed mean/std for Gaussian robustness. ATR measures whether same-state clean/noisy rollout-disagreement tails contract at recovered endpoints, and SMPR checks whether task-grounded near-boundary separations survive that contraction. Unseen-stressor scores are auxiliary scope checks rather than additional core diagnostics.

## 5 Experiments

The experiments use additive Gaussian pixel noise as the primary controlled stressor for a representation-diagnostic question. We first show that the stressor changes closed-loop behavior. We then ask where recovery appears in the model, prioritizing cross-task ACPC/paired-rollout evidence and keeping task-specific representation measurements only as discriminability checks. The study is a matched-stressor trace through the encoder–predictor–planner chain; point-best noise-level ranking, external method comparison, and broad perturbation-family transfer are outside its scope.

All results use the protocol in Section 4: LeWM-base denotes the no-noise checkpoint, and LeWM+noise denotes the eight-level  $\sigma_{\max}$  sweep.

### 5.1 JEPA control fragility under Gaussian observation noise

Table 2 reports LeWM-base success rates under unperturbed and noised evaluation. Each training-seed value is first averaged over the three evaluation seeds; the table reports mean  $\pm$  population std across the three training seeds.

**Table 2:** LeWM-base under test-time Gaussian-noise settings. Values are mean  $\pm$  population std across three training seeds; each training-seed value averages three evaluation seeds  $\times$  100 trajectories. Drop is computed per training seed as clean success minus observation-noise 0.08 success, then summarized.

| Task    | unpert.          | obs 0.05         | obs 0.08         | obs+goal 0.08    | drop             |
|---------|------------------|------------------|------------------|------------------|------------------|
| TwoRoom | 94.67 $\pm$ 0.72 | 81.33 $\pm$ 6.77 | 68.78 $\pm$ 2.45 | 53.56 $\pm$ 2.57 | 25.89 $\pm$ 2.38 |
| PushT   | 81.78 $\pm$ 4.89 | 19.22 $\pm$ 9.12 | 7.22 $\pm$ 5.81  | 4.89 $\pm$ 1.64  | 74.56 $\pm$ 5.55 |
| Reacher | 59.22 $\pm$ 1.03 | 33.56 $\pm$ 1.77 | 18.22 $\pm$ 0.42 | 13.78 $\pm$ 1.50 | 41.00 $\pm$ 1.44 |
| Cube    | 65.22 $\pm$ 1.10 | 46.67 $\pm$ 1.96 | 43.11 $\pm$ 3.27 | 45.78 $\pm$ 0.79 | 22.11 $\pm$ 2.78 |

LeWM-base is strong on unperturbed images, but observation noise alone already produces large closed-loop drops: PushT loses 74.56  $\pm$  5.55 pts at observation-noise 0.08, Reacher 41.00  $\pm$  1.44 pts, TwoRoom 25.89  $\pm$  2.38 pts, and Cube 22.11  $\pm$  2.78 pts. The observation+goal column shows the stronger full-input-stream Gaussian-noise stress case; it further degrades TwoRoom and Reacher, but is not the primary robustness endpoint because the goal image is also perturbed. The drop pattern across tasks remains informative: PushT degrades most sharply, consistent with contact-heavy continuous control being sensitive to visual precision.

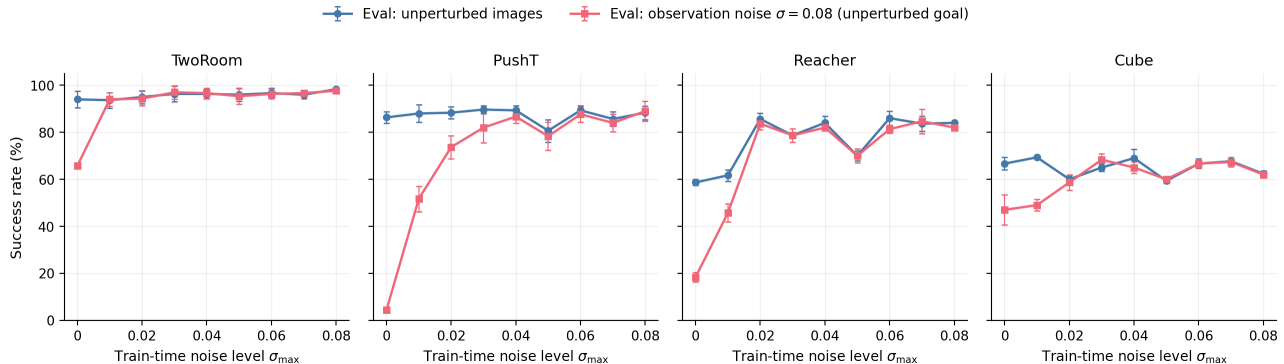
### 5.2 Noise augmentation supplies recovered checkpoints for diagnosis

Table 3 summarizes the sweep points needed for the main argument. All entries use the unified  $n = 3$  evaluation seeds (42/43/44)  $\times$  100 trajectories protocol.



**Table 3:** LeWM noise-sweep summary. Values are success rate mean  $\pm$  population std across evaluation seeds. The two point-best columns may refer to different training-noise levels; they summarize the sweep envelope rather than a point-optimal checkpoint claim.

| Task    | unpert. base     | obs 0.08 base    | best clean row          | best obs0.08 row        | Main reading                              |
|---------|------------------|------------------|-------------------------|-------------------------|-------------------------------------------|
| TwoRoom | 94.00 $\pm$ 3.56 | 65.67 $\pm$ 1.25 | 98.33 $\pm$ 0.47 (0.08) | 97.67 $\pm$ 0.94 (0.08) | large recovery; visually redundant regime |
| PushT   | 86.33 $\pm$ 2.36 | 4.33 $\pm$ 1.25  | 89.67 $\pm$ 1.70 (0.03) | 89.00 $\pm$ 4.32 (0.08) | large recovery; contact-sensitive state   |
| Reacher | 58.67 $\pm$ 1.25 | 18.33 $\pm$ 2.05 | 86.00 $\pm$ 2.94 (0.06) | 84.67 $\pm$ 5.19 (0.07) | recovery plus fixed-epoch clean lift      |
| Cube    | 66.67 $\pm$ 2.62 | 47.00 $\pm$ 6.38 | 69.33 $\pm$ 0.47 (0.01) | 68.33 $\pm$ 2.49 (0.03) | weaker recovery signal                    |



**Figure 1:** Noise-training sweep across four tasks. Blue circles: unperturbed evaluation; red squares: observation-only Gaussian noise  $\sigma = 0.08$  with an unperturbed goal. Error bars show the population standard deviation across the three evaluation seeds. The curves document broad task-dependent recovery plateaus rather than point-optimal ranking.

The sweep contributes two observations to the main argument. First, input-side noise can be an effective intervention in this protocol: PushT recovers from 4.33% to 89.00% at observation-noise 0.08, and Reacher from 18.33% to 84.67%. Second, the grid supplies representative recovered checkpoints for the downstream diagnostic question: what changed in the encoder–predictor–planner chain?

Reacher also shows a fixed-epoch clean-control lift. Noise training improves not only observation-noise robustness but also unperturbed control: the representative clean row rises from 58.67% at the no-noise baseline to about 86% under noise training. We interpret this as a fixed-protocol regularization or stabilization effect rather than pure robustness recovery. The three-training-seed replication below makes the endpoint less dependent on seed 3072, but longer learning curves would still be needed to separate regularization from optimization timing.

The completed Gaussian training-seed replication strengthens the closed-loop part of the claim. It repeats the full Gaussian sweep for LeWM seeds 3073 and 3074 and combines them with the canonical seed-3072 sweep. Table 4 summarizes the observation-only Gaussian endpoint across all three training seeds.

**Table 4:** Three-training-seed Gaussian replication for LeWM. Values are mean  $\pm$  population std across training seeds 3072/3073/3074; each seed point is itself the mean over evaluation seeds 42/43/44 with 100 trajectories per eval seed. The endpoint is observation-only Gaussian evaluation at 0.08 with the goal image unperturbed. Regret is best obs 0.08 minus the fixed  $\sigma_{\max} = 0.08$  endpoint; the best- $\sigma_{\max}$  range shows the task/seed-dependent plateau.

| Task    | base obs 0.08    | std0.08 obs 0.08 | gain             | best obs 0.08    | regret          | best $\sigma_{\max}$ range |
|---------|------------------|------------------|------------------|------------------|-----------------|----------------------------|
| TwoRoom | 68.78 $\pm$ 2.46 | 97.11 $\pm$ 0.57 | 28.33 $\pm$ 2.68 | 97.67 $\pm$ 0.27 | 0.56 $\pm$ 0.79 | 0.05–0.08                  |
| PushT   | 7.22 $\pm$ 5.81  | 85.78 $\pm$ 3.45 | 78.56 $\pm$ 5.18 | 87.67 $\pm$ 0.98 | 1.89 $\pm$ 2.67 | 0.04–0.08                  |
| Reacher | 18.22 $\pm$ 0.42 | 81.55 $\pm$ 0.88 | 63.34 $\pm$ 1.25 | 83.78 $\pm$ 0.63 | 2.22 $\pm$ 0.87 | 0.05–0.07                  |
| Cube    | 43.11 $\pm$ 3.27 | 62.56 $\pm$ 1.55 | 19.45 $\pm$ 4.53 | 66.44 $\pm$ 1.34 | 3.89 $\pm$ 2.20 | 0.03–0.05                  |

Across three training seeds, the no-noise checkpoints retain large observation-noise cliffs, while the  $\sigma_{\max} = 0.08$  endpoint is within 0.56–3.89 pp of each task’s best obs0.08 row. PushT remains the sharpest stress case, improving by 78.56  $\pm$  5.18 points at the fixed high-noise endpoint; Reacher improves by 63.34  $\pm$  1.25 points. The

nonzero regret and task-dependent best- $\sigma_{\max}$  ranges support plateau wording rather than a universal optimum at exactly  $\sigma_{\max} = 0.08$ .

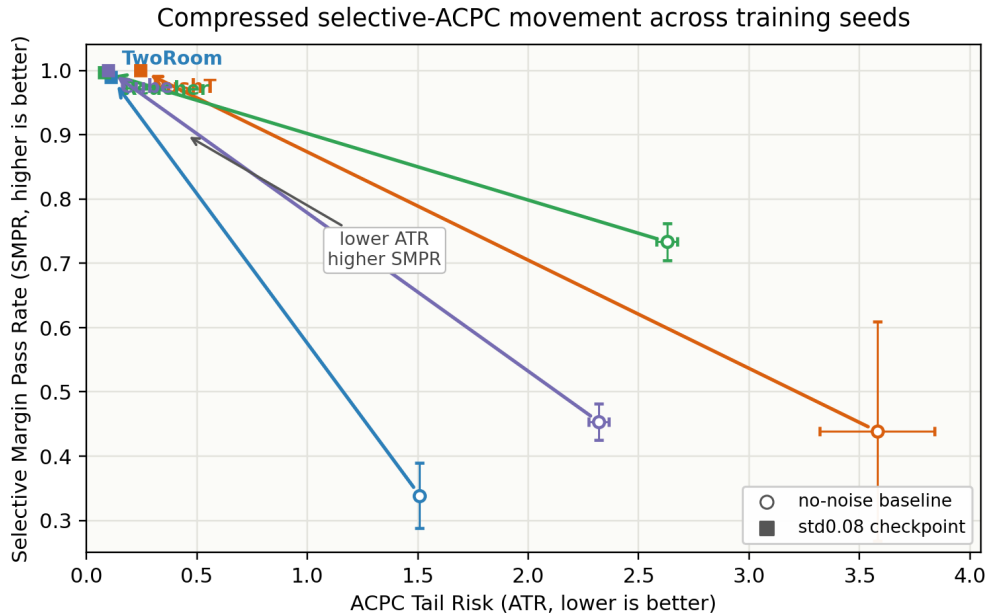
A three-seed blur/resize score check is reported in Appendix C. It is an auxiliary severe-stressor check, not part of the matched-Gaussian claim. The next subsection asks whether the recovered Gaussian endpoints also move in the two compressed diagnostic readouts.

### 5.3 Compressed selective-ACPC diagnostics

For the paper-facing diagnostic, ATR is the 90th percentile of same-state clean/noisy ACPC-H rollout distance normalized by the clean transition scale. SMPR is the task-grounded near-boundary selective-margin pass rate at margin zero. Both are computed without retraining on the fixed no-noise and  $\sigma_{\max} = 0.08$  checkpoints for training seeds 3072/3073/3074. Table 5 reports population mean/std across training seeds.

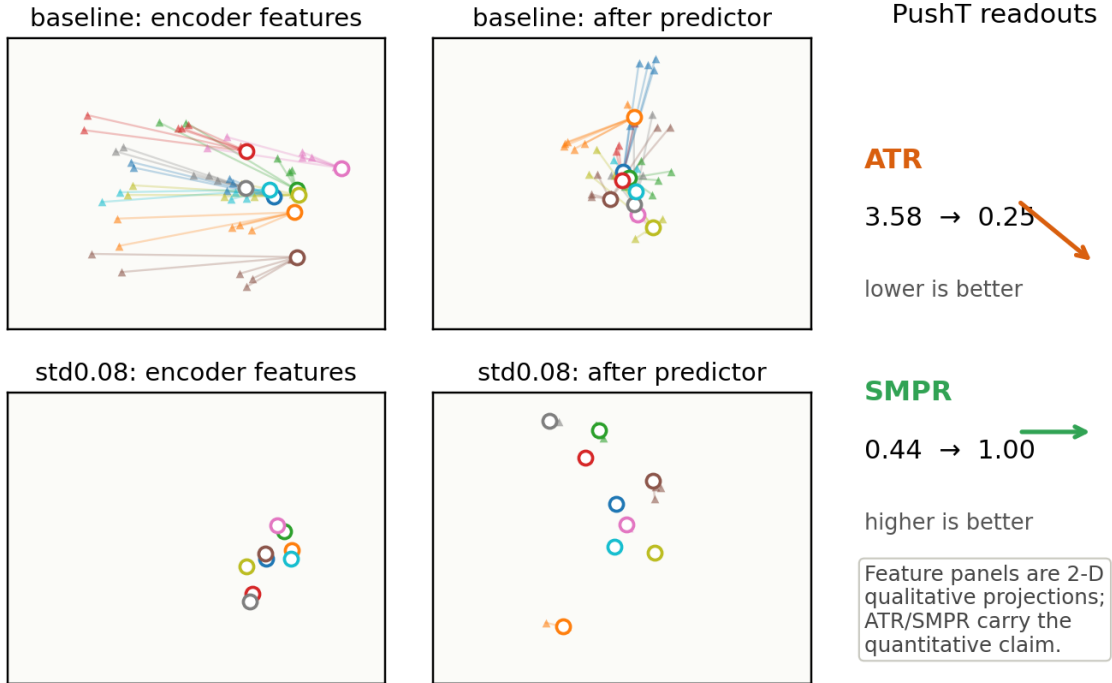
**Table 5:** Compressed selective-ACPC diagnostics over training seeds 3072/3073/3074. ATR is the q90 normalized same-state clean/noisy action-conditioned rollout disagreement; lower is better. SMPR is the task-grounded near-boundary selective-margin pass rate; higher is better.

| Task    | ATR base          | ATR std0.08       | SMPR base       | SMPR std0.08    |
|---------|-------------------|-------------------|-----------------|-----------------|
| TwoRoom | $1.509 \pm 0.010$ | $0.111 \pm 0.016$ | $0.34 \pm 0.05$ | $0.99 \pm 0.01$ |
| PushT   | $3.580 \pm 0.260$ | $0.247 \pm 0.017$ | $0.44 \pm 0.17$ | $1.00 \pm 0.00$ |
| Reacher | $2.628 \pm 0.048$ | $0.082 \pm 0.005$ | $0.73 \pm 0.03$ | $1.00 \pm 0.00$ |
| Cube    | $2.320 \pm 0.045$ | $0.100 \pm 0.007$ | $0.45 \pm 0.03$ | $1.00 \pm 0.00$ |



**Figure 2:** Compressed selective-ACPC diagnostic plane. Each arrow moves from the no-noise baseline to the fixed  $\sigma_{\max} = 0.08$  checkpoint; lower ATR and higher SMPR are the intended diagnostic direction. Error bars show population std across training seeds.

### PushT same-state feature neighborhoods under visual perturbations



Circles are clean anchors; triangles are same-state noisy views. Shared colors indicate the same underlying state.

**Figure 3:** Qualitative PushT feature-neighborhood illustration for the same no-noise versus  $\sigma_{\max} = 0.08$  contrast. Circles mark clean anchors and triangles mark same-state noisy views after a shared 2-D projection of encoder features or post-predictor rollout features. The panel is visual intuition only; ATR and SMPR remain the quantitative paper-facing readouts.

The two readouts move in the intended selective direction on all four tasks: recovered endpoints have much smaller same-state predictive tails and much higher task-grounded margin pass rates. Figure 2 visualizes the same movement as Table 5; it is not an additional diagnostic or a point-ranking rule inside a plateau. Figure 3 provides a qualitative feature-neighborhood illustration of the same intuition on PushT: same-state noisy views become tighter after noise training, while the quantitative claim still rests on ATR, SMPR, and closed-loop behavior. Cube remains the useful boundary case: the compressed diagnostics improve strongly, but the closed-loop Gaussian gain and regret in Table 4 are weaker than PushT or Reacher, so closed-loop evaluation remains the behavioral authority.

SMPR uses programmatic task-state proxy labels, not oracle semantic annotations: TwoRoom uses doorway/room-side labels, PushT uses T-block pose cells, Reacher uses target/end-effector relation bins, and Cube uses cube-pose/goal-relation cells. Each anchor selects the closest label-crossing neighbor inside the closest 35% state-distance neighborhood and passes if that clean different-state rollout distance exceeds the same-state noisy radius. The no-noise baselines are weak under this stricter audit, while the high-noise endpoint reaches near-one pass rate; stronger hand-labeled or simulator-derived contact, topology, action-value, or cost-to-go labels remain future validation.

#### 5.4 Bounded unseen-stressor check

The main claim remains matched Gaussian-noise robustness. After freezing the Gaussian grid, we ran strongest-severity unseen-stressor score checks to test whether the fixed  $\sigma_{\max} = 0.08$  endpoint transfers outside the training perturbation family. Table 6 reports behavior only. TwoRoom and Reacher improve under strongest blur, whereas PushT and Cube do not show a clear resize gain at the scale of training-seed and evaluation variance. We use this as bounded behavior outside the matched Gaussian setting, not as a universal cross-perturbation robustness claim.

**Table 6:** Bounded unseen-stressor score check. Values are mean  $\pm$  population std across LeWM training seeds 3072/3073/3074; each seed point averages evaluation seeds 42/43/44 from strongest-severity unseen runs. Stress  $\Delta$  is the fixed  $\sigma_{\max} = 0.08$  checkpoint minus the no-noise baseline.

| Task    | stress        | baseline stress   | std0.08 stress   | stress $\Delta$   | reading          |
|---------|---------------|-------------------|------------------|-------------------|------------------|
| TwoRoom | blur $k = 15$ | $47.67 \pm 5.44$  | $90.78 \pm 5.38$ | $+43.11 \pm 8.90$ | clear score gain |
| Reacher | blur $k = 15$ | $22.00 \pm 3.78$  | $71.22 \pm 1.10$ | $+49.22 \pm 3.29$ | clear score gain |
| PushT   | resize 0.25   | $63.44 \pm 14.05$ | $66.33 \pm 8.38$ | $+2.89 \pm 17.98$ | no clear gain    |
| Cube    | resize 0.25   | $57.00 \pm 1.96$  | $56.11 \pm 0.57$ | $-0.89 \pm 1.40$  | no clear gain    |

## 6 Discussion and limitations

**What the evidence supports.** This is a controlled diagnostic study, not a method paper. Closed-loop evaluation establishes the behavior; ATR checks whether same-state noisy views have small action-conditioned predictive tails; SMPR checks whether task-grounded near-boundary distinctions survive that contraction. The primary statistic is the full three-training-seed Gaussian replication, with evaluation-seed variance kept as the within-seed measurement scale. These standard deviations are descriptive over three training seeds, not asymptotic significance claims.

**Scope.** Gaussian pixel noise remains the training/evaluation axis. The blur/resize rows are bounded behavior outside the matched Gaussian setting, not a transfer theorem. ATR/SMPR are not a point-best  $\sigma_{\max}$  predictor and do not replace closed-loop evaluation inside a broad plateau. We do not claim superiority over DrQ-style augmentation, TD-MPC2, DreamerV3, or robust MPC methods; the contribution is a no-retraining diagnostic for JEPA latent world-model checkpoints under a controlled Gaussian stressor.

**What remains.** The results argue against a too-strong reading of latent prediction: future-latent prediction alone does not guarantee closed-loop visual robustness. Stronger discriminability evidence would require hand-labeled or simulator-derived contact, topology, action-value, or cost-to-go labels, especially for PushT and Cube. Future methods can turn ATR/SMPR into objectives, but this paper keeps that as a separate method-paper path after diagnostic validity is established.

## 7 Conclusion

This paper frames Gaussian-noise robustness in JEPA latent world-model control as a selective diagnostic problem. Closed-loop behavior establishes the phenomenon; ATR measures same-state predictive-stability tails; SMPR checks task-grounded anti-collapse margins. On fixed LeWM checkpoints, input-side noise training recovers broad matched-Gaussian robustness plateaus across three training seeds, and the two compressed diagnostics move in the expected direction at recovered endpoints.

The theory is deliberately diagnostic: sampled-pool ACPC exposes why tail risk matters and why clean margins prevent closed-loop guarantees, while local Gaussian linearization connects the measured tail to action-conditioned encoder–predictor sensitivity. The broader implication is not solved generalization; it is a concrete audit target for future methods: stable action-conditioned predictive dynamics under nuisance variation, while preserving distinctions needed for planning.

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## A Proofs and Calibration for ACPC Diagnostics

**Reading.** These proofs calibrate ATR and SMPR as diagnostics; they do not provide closed-loop control guarantees.

*Proof of Proposition 2.* For any fixed action sequence  $\mathbf{a}$ , the assumption gives  $d_H(U_h(\mathbf{a}), U_{\tilde{h}}(\mathbf{a})) \leq \epsilon$ . The cost readout  $J$  is  $L_J$ -Lipschitz on the rollout-readout sequence, so  $|C_h(\mathbf{a}, g) - C_{\tilde{h}}(\mathbf{a}, g)| \leq L_J\epsilon$ . Taking the maximum over the fixed candidate set gives the claim.  $\square$

*Proof of Proposition 3.* Let  $\mathbf{a}^*$  be the clean best candidate and  $\mathbf{a}^{(2)}$  the clean runner-up. If every candidate cost moves by at most  $L_J\epsilon$  and the clean margin is larger than  $2L_J\epsilon$ , then the perturbed cost of  $\mathbf{a}^*$  remains below the perturbed cost of every other candidate. Thus the top candidate is unchanged.  $\square$

*Proof of Theorem 1.* For a sampled pool of size  $K$ , the probability that at least one candidate has clean/noisy rollout discrepancy above  $\epsilon$  is at most  $K\delta$  by the union bound. On the complement of that event, every candidate cost moves by at most  $L_J\epsilon$ . A flip can then occur only if the clean top margin is at most  $2L_J\epsilon$ , which contributes the margin-tail term. This proves the stated sampled-pool instability bound.  $\square$

**Finite-sample calibration.** For a bounded Bernoulli diagnostic with empirical pass rate  $\hat{p}$  over  $n$  fixed pairs, Hoeffding gives

$$\Pr \left[ p < \hat{p} - \sqrt{\frac{\log(1/\alpha)}{2n}} \right] \leq \alpha. \quad (9)$$

We use this only as calibration for the fixed SMPR pair construction; nearby dataset windows can be dependent.

**Local Gaussian sensitivity.** For small Gaussian pixel noise  $\xi$ , a first-order expansion gives  $E_\theta(o + \xi) = E_\theta(o) + J_E(o)\xi + \mathcal{R}(\xi)$  with  $\|\mathcal{R}(\xi)\| = O(\|\xi\|^2)$ . Composing the encoder with the action-conditioned rollout map shows that same-state rollout disagreement measures local sensitivity of the encoder–predictor composition, not encoder invariance alone. ATR reports a high-quantile version of that paired rollout disagreement.

## B Experimental Details

**Reading.** These details define the fixed evaluation and diagnostic protocol used by the main tables.

LeWM is evaluated on PushT, TwoRoom, Reacher, and Cube using the canonical three evaluation seeds 42/43/44 and 100 trajectories per seed. The main Gaussian robustness endpoint perturbs observation pixels with Gaussian noise at standard deviation 0.08 while keeping the goal image clean. Noise-trained checkpoints use full-sequence input-side Gaussian augmentation with  $\sigma_{\max} \in \{0.01, \dots, 0.08\}$ .

ATR is computed at each task–training-seed–checkpoint row as the 90th percentile same-state ACPC-H distance normalized by the clean transition scale. SMPR uses task-grounded near-boundary proxy labels at margin zero. The paper reports population mean/std across training seeds 3072/3073/3074.

## C Unseen-Stressor Score Check

**Reading.** This appendix reports behavior outside the matched Gaussian family without expanding the paper’s main claim.

After the Gaussian grid and training-seed replication were frozen, we ran strongest-severity unseen-stressor score checks on all three LeWM training seeds, comparing no-noise training to  $\sigma_{\max} = 0.08$  under Gaussian blur with kernel size 15 and resize/down-upsample with factor 0.25. This appendix reports behavior only; it does not broaden the main claim into cross-perturbation robustness.

**Table 7:** Three-training-seed unseen-stressor score check. Values are mean  $\pm$  population std across LeWM training seeds 3072/3073/3074. Stress  $\Delta$  is  $\sigma_{\max} = 0.08$  stress success minus no-noise stress success.

| Task    | unseen stress | baseline stress   | std0.08 stress   | stress $\Delta$   | drop impr.        | reading          |
|---------|---------------|-------------------|------------------|-------------------|-------------------|------------------|
| TwoRoom | blur $k = 15$ | $47.67 \pm 5.44$  | $90.78 \pm 5.38$ | $+43.11 \pm 8.90$ | $+40.89 \pm 8.23$ | clear score gain |
| Reacher | blur $k = 15$ | $22.00 \pm 3.78$  | $71.22 \pm 1.10$ | $+49.22 \pm 3.29$ | $+30.22 \pm 4.16$ | clear score gain |
| PushT   | resize 0.25   | $63.44 \pm 14.05$ | $66.33 \pm 8.38$ | $+2.89 \pm 17.98$ | $-3.78 \pm 15.56$ | no clear gain    |
| Cube    | resize 0.25   | $57.00 \pm 1.96$  | $56.11 \pm 0.57$ | $-0.89 \pm 1.40$  | $+2.78 \pm 3.40$  | no clear gain    |